Nicholas Domke

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EDUCATION

Louisiana State University (LSU), Baton Rouge, LA **May 2027**
Bachelor of Science in Mechanical Engineering
Minors: Robotics, Computer Science (in progress)
Related Coursework: Thermodynamics, Heat Transfer, Fluid Dynamics, Manufacturing Processes, System Dynamics, Simulation Methods, Statics, Strength of Materials, Circuits, Electronics, Kinematics, Introduction to Robotics, Welding Engineering 1, Industrial Robotics, Instrumentation and Measurements, Machine Design

Graduate Study Intent
Applying to Master of Science programs in Controls and Mechanical Systems
Applications for Fall 2027

EXPERIENCE

Mechanical Engineering Intern **May 2026 – August 2026**
Integer Technologies, Baton Rouge, Louisiana
- Designed 3D-printed mounts in Fusion 360 to integrate an acoustic modem, housed in a Blue Robotics watertight enclosure, onto a BlueROV2-based vehicle (BlueLink SARbot) for in-water testing
- Developed 4 design concepts, starting from a modular rail mount, then printed and refined 5 to 6 iterations of the final mount in PETG at 100% infill
- Designed a Raspberry Pi mount that fits inside the watertight enclosure, refined over 6 versions and printed in PLA
- Selected quick-connect bulkheads and cables so the modem connects and disconnects while the vehicle is in the water; the mounts performed well in pool and lake tests

Student Researcher **March 2026 – Present**
LSU iCORE Lab, Baton Rouge, Louisiana
- Building a path planner for unmanned surface vessels in simulation, combining artificial potential field and ant colony optimization models to avoid static and moving obstacles
- Designed a SolidWorks mount that holds a laser and a depth sensor on a UR5e robot arm to detect surface imperfections on wind turbine blades
- Designed the end effector for a 3D-printed 5-DOF mock welding arm and measured 0.83% position error over 5 straight-line runs

Math Tutor **August 2025 – May 2026**
LSU Math Lab, Baton Rouge, Louisiana
- Tutored up to 100 students in college algebra and trigonometry to improve critical thinking and comprehension

Volunteer Referee **2017 – 2022**
FIRST LEGO League
- Refereed competitions and advised teams on how to improve before their next run

Server **May 2025 – January 2026**
Enterrium, Schaumburg, Illinois
- Maintained an enthusiastic attitude while managing multiple tasks for customer service

Server **May 2023 – August 2024**
Texas Roadhouse, Mount Prospect, Illinois
- Maintained an enthusiastic attitude while managing multiple tasks for customer service

SKILLS

- Software: SolidWorks (8 years), Onshape, Fusion 360, MATLAB, Python, Robotic Operating System 2 (ROS 2), Arduino IDE, VCarve
- Hardware: Manual Lathe, Haas CNC Router, CNC Router, MIG Welding, 3D Printing (PLA, PETG, Markforged Onyx), Soldering, Drill Press, Miter Saw, Band Saw

PROJECTS

Senior Design: Electric Boat **August 2026 – May 2027**

- Team lead for a team of 8 (4 mechanical, 3 electrical, 1 computer engineer) building an electric boat for ASNE's Promoting Electric Propulsion (PEP) national student competition, sponsored by ONR
- Designing the custom, jet-ski-inspired hull in SolidWorks
- Leading the first LSU senior design team to take on this project, designing a new boat on a $7,000 first-year budget

Mock Welding Robotic Arm **March 2026 – May 2026**

- Designed the end effector in SolidWorks for a 5-DOF DIY robot arm, on a team of 3 (two mechanical engineers, one computer science student)
- 3D printed the arm and end effector, and soldered wires to the servo potentiometers so the arm could read its joint positions
- Verified the CAD model in simulation software, then measured the real arm: 0.83% position error over 5 straight-line runs

Robot Maze Navigation Semester Project for Introduction to Robotics **October 2025 – December 2025**

- Designed a mobile robot base and camera system in SolidWorks and 3D printed them, on a team of 3 (two mechanical engineers, one computer science student)
- The robot autonomously navigated and finished the maze using its camera

Manufacturing Lab Projects **January 2025 – April 2025**

- Operated a manual lathe to machine a clutch alignment pin to a tolerance of ±0.005 inches
- MIG welded a steel stool
- Programmed and operated a Haas CNC router to cut an aluminum heat exchanger, writing the G-code and tool paths

Inclusive Playground, Sophomore Design Course **September 2023 – December 2023**

- Researched existing playgrounds with a team of 6 mechanical engineers and designed one more inclusive for all types of disabilities, following ADA guidelines
- Designed the playground model in SolidWorks and combined each team member's parts into the final assembly
- Chose aluminum and polyethylene to keep the playground lightweight for easier construction

WildStang Robotics Team, FIRST Robotics Competition **August 2018 – May 2022**

- Primary team lead junior and senior years (co-lead sophomore year) and a driver all four years, on a team of about 80
- Won 4 of the 5 regional competitions attended, plus an at-home challenge during COVID
- Led design, manufacturing, and assembly of 5 robots over 4 years, including designing the drive base and gearbox; 95% of the 2020 robot was manufactured in house
- Led CAD design using SolidWorks and Onshape while teaching younger team members CAD and power tools (hand lathes, drills, miter saws, band saws)
- Managed project timelines and coordinated with each sub-team so each robot was working within the 8-week build season
- Ran the team's CNC router using VCarve and 3D printed Onyx parts on Markforged 3D printers
- Designed, modeled, and assembled a new workshop storage system in Onshape so essentials could be moved easily for competitions